A Complete Treatment of Budget Pacing: From Single-Advertiser Closed Form to Market Equilibrium

Robert Hou

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Abstract

This note derives, from first principles, the optimization theory of **budget pacing** in online advertising. Starting from a single advertiser's stochastic spend objective, we rigorously obtain its closed-form optimal policy in the **fluid limit**, and reveal that the **pacing multiplier** is the **Lagrangian dual variable** (shadow price) of the budget constraint. We then extend the model to multiple advertisers, introduce platform revenue, and obtain a two-sided market whose steady state is a **competitive equilibrium**—in which the game among advertisers is a **Nash / pacing equilibrium** and the platform, as mechanism designer, induces a **Stackelberg** structure. Finally we unify three viewpoints—stochastic dynamic programming, reinforcement learning / stochastic approximation, and PI feedback control—and characterize the boundary at which closed-form solutions exist.

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1 Introduction: Advertising as Constrained Optimization

Search, recommendation, and advertising are mathematically **one and the same problem**: constrained optimization over a vast candidate space. The reason engineering decomposes them into a retrieval $\rightarrow$ ranking pipeline is that an exact arg max over the full corpus is infeasible—retrieval shrinks the candidate space from billions to thousands, and ranking approximates the optimal allocation within that small space.

Unifying view. All three systems are "constrained optimization over a candidate space," and the retrieval–ranking pipeline is its **hierarchical approximate solution**:

- **Search**: objective = maximize user satisfaction for the query; the constraint is mainly relevance.
- **Recommendation**: objective = maximize retention / dwell time.
- **Advertising**: objective = platform revenue, **subject to each advertiser's budget constraint**, with multiple agents competing in the same auction.

What advertising adds over search is precisely two layers: the **budget constraint** (whose dual variable is the pacing multiplier) and the **auction game** (market equilibrium). Search is the special case of advertising with the budget and auction removed.

This note focuses on advertising, the most complex case, and carries the derivation all the way through.

2 Two Often-Confused Notions: Cap vs. Pacing

Budget Cap — a correctness constraint (hard). Actual spend must not significantly exceed the budget. A violation = overcharging the advertiser = a financial/legal incident. Its SLA is **bounded overspend**, not zero overspend.

Pacing — an optimization objective (soft). Spend the budget smoothly and on rhythm throughout the day, so it lands on high-value windows rather than cheap, low-quality early-morning traffic. A pacing failure = money spent suboptimally, but it **never overspends** (that is the cap's job). Its SLA is statistical (stay near the target curve most of the time).

Their failure modes form a pair: **overspend** (breaching the cap) and **under-delivery** (budget left unspent, the opportunity window evaporating). Leaving a buffer against overspend pushes up under-delivery; the two trade off against each other. The optimization theory below characterizes **pacing** (the soft objective); the cap is enforced by an independent hard layer (local-quota self-stop) outside this framework.

3 Single-Advertiser Model

3.1 eCPM and Billing

When an impression opportunity arrives, each candidate ad's **effective cost per mille** eCPM is computed on the spot:

$$\text{eCPM} = 1000 \times \underbrace{\text{bid} \times p(\text{conv})}_{\text{per-impression value } v},$$

where $p(\text{conv}) = p_{\text{CTR}} \cdot p_{\text{CVR}}$ is model-estimated. The constant 1000 cancels in ranking, so we write the per-impression value as

$$v = \text{bid} \cdot p_{\text{CTR}} \cdot p_{\text{CVR}}.$$

Under oCPM the billable event is the **impression** (high frequency), but the bid is attached to **conversion**—to maximize its own eCPM the platform must serve impressions to high-conversion users, which (when p is well calibrated) aligns its incentives with the advertiser's.

3.2 Decisions and Notation

Discretize the day $[0, T]$ into windows $k = 1, \dots, K$, each of length Δt.

Symbol	Meaning
B	daily budget
$S^*(t),\ S_k^*$	target cumulative-spend curve (estimated from history)
$S(t),\ S_k$	actual cumulative spend
$\mu_k \geq 0$	pacing multiplier (decision variable, applied to the bid)
N_k	impression arrival rate in window k (per second)
$M_k = N_k \Delta t$	number of impression opportunities in window k
v_k	impression value (eCPM)
c_k	charge upon winning one impression
p_k	win rate $= P(\mu_k v_k > \text{auction}_k)$

Serving rule: the ad competes with paced bid $\mu_k v_k$; if it wins, it is shown and charged. Larger μ means stronger competitiveness and faster spend.

4 Offline Optimum: Knapsack and the Dual Variable

4.1 Knapsack Formulation

With an oracle (all opportunities of the day known), the advertiser maximizes total value won within budget:

$$\max_{x_i \in \{0,1\}} \sum_{i=1}^{N} v_i x_i \quad \text{s.t.} \quad \sum_{i=1}^{N} c_i x_i \leq B.$$

This is formally a 0/1 knapsack (NP-hard). But in advertising the unit cost is **infinitesimal** relative to the budget ($c_i/B \sim 10^{-7}$) and the number of opportunities **enormous**, so the integrality gap collapses.

Proposition 1 (Integrality gap)**.** *Let the LP relaxation allow* $x_i \in [0, 1]$. *Then*

$$0 \leq \text{OPT}_{\text{LP}} - \text{OPT}_{\text{IP}} \leq v_{\max}.$$

Proof sketch. Put the problem in standard form with slacks: budget slack s_0 and upper-bound slacks $x_i + s_i = 1$. There are $2N{+}1$ variables and $N{+}1$ equality constraints. At a basic feasible solution (BFS), non-basic variables are 0 and exactly $N{+}1$ variables are basic. A fractional $x_k \in (0, 1)$ requires both x_k

and s_k to be basic (two basis slots); when the budget is exhausted $s_0 = 0$ is non-basic, and counting basis slots shows **at most one fractional variable**. Rounding that single fractional variable down to 0 yields an integer feasible solution losing $< v_k \le v_{\max}$. □ □

Since $v_{\max}/\mathrm{OPT} \sim 10^{-7}$, one may safely treat the problem as continuous (fractional knapsack).

4.2 Greedy Optimality and the Threshold

Proposition 2 (v/c greedy is optimal)**.** *The optimum of fractional knapsack is: include items in decreasing order of bang-per-buck v_i/c_i until the budget is exhausted; exactly one critical item is split.*

Exchange argument. Suppose an optimum x^* contains a, b with $\frac{v_a}{c_a} > \frac{v_b}{c_b}$ but $x_a^* < 1$ and $x_b^* > 0$. Move a budget increment δ from b to a: spend is conserved while value changes by $\delta(\frac{v_a}{c_a} - \frac{v_b}{c_b}) > 0$, a strict improvement—contradiction. Hence the optimum fills items in decreasing bang-per-buck. □ □

Equivalently, there exists a threshold λ such that the optimal policy is

$$\text{include } i \iff \frac{v_i}{c_i} \ge \lambda, \qquad \lambda \text{ chosen so spend equals } B.$$

This λ is exactly the bang-per-buck v_k/c_k of the critical item.

Core result. This threshold λ is the **Lagrangian dual variable** of the budget constraint, i.e. the budget's **shadow price**—the opportunity cost of one more dollar spent. With the Lagrangian

$$\mathcal{L} = \sum_i v_i x_i - \lambda\Big(\sum_i c_i x_i - B\Big),$$

the per-impression optimal decision becomes "include iff $v_i - \lambda c_i > 0$." In the simplex method $\lambda^\top = c_B^\top B^{-1}$ emerges implicitly from the basis change; under a single-resource knapsack it scalar-collapses to $\lambda = v_k/c_k$. **The pacing multiplier μ is the real-time embodiment of this λ: tight budget $\Rightarrow$ λ large $\Rightarrow$ raise the admission bar; loose budget $\Rightarrow$ λ small $\Rightarrow$ open up.**

The offline λ needs an oracle; online, the future is unknown, so λ must be **estimated in real time**. This turns a static optimization into online control—the through-line of the rest of this note.

5 The Full Stochastic Objective

In a real system, spend is stochastic. We measure pacing quality by **normalized deviation** (absolute deviation is not comparable: being off by \$100 means wildly different things for a \$200 vs. a \$2M budget):

$$\min_{\{\mu_k\}} \sum_n \frac{1}{(S_n^*)^2}\,\mathbb{E}\Big[\big(S_n - S_n^*\big)^2\Big] \quad \text{s.t.} \quad S_T \le B,\ \ \mu_k \ge 0.$$

By the bias–variance decomposition:

$$\mathbb{E}\big[(S_n - S_n^*)^2\big] = \underbrace{\big(\mathbb{E}[S_n] - S_n^*\big)^2}_{\text{bias}^2\ (\mu\text{-controllable})} + \underbrace{\mathrm{Var}(S_n)}_{\text{stochastic floor } (\mu\text{-uncontrollable})}.$$

The controller can only reduce bias^2; $\mathrm{Var}(S_n)$ stems from the inherent randomness of auctions/traffic and is a floor μ cannot remove. **This explains why pacing never sticks perfectly to the curve**: even with the mean perfectly aligned, variance makes S_n jitter.

Expected cumulative spend:

$$\mathbb{E}[S_n] = \sum_{k=1}^{n} N_k \Delta t \, p_k \, c_k, \qquad p_k = P(\mu_{k-1} v_k > \text{auction}_k).$$

6 Variance Expansion and the Fluid Limit

6.1 Per-Step Variance

Window-k spend is $\Delta S_k = \sum_{m=1}^{M_k} c_{k,m} \nVdash_{k,m}$, with two layers of randomness: whether one wins, $\nVdash_{k,m}$ (Bernoulli), and the winning charge $c_{k,m}$.

Assumption 1 (First-price simplification)**.** *Winning charges a deterministic value c_k tied to one's own bid (first-price / known charge), removing the randomness of the second-highest price. Historically ad billing was first-price, then GSP was introduced to stabilize bidding games; here we take first-price for a closed form.*

Assumption 2 (Within-window i.i.d.)**.** *Within a small window, impressions are independent and the win rate p_k is approximately constant.*

Then $\sum_m \nVdash_{k,m} \sim \text{Binomial}(M_k, p_k)$, and

$$\text{Var}(\Delta S_k) = c_k^2 \cdot M_k \, p_k (1 - p_k) = c_k^2 \, N_k \Delta t \, p_k (1 - p_k).$$

Windows being independent, $\text{Var}(S_n) = \sum_{k \le n} c_k^2 N_k \Delta t \, p_k (1 - p_k)$.

6.2 Law of Large Numbers Kills Variance

Compare term by term the mean and variance contributions:

$$\underbrace{N_k \Delta t \, c_k \, p_k}_{A_n \text{ term}} \qquad \text{vs.} \qquad \underbrace{N_k \Delta t \, c_k^2 \, p_k (1 - p_k)}_{V_n \text{ term}}.$$

The variance term carries an extra factor c_k ($c_k \sim 0.006$) and is only a first-order sum in M_k, whereas the bias term is the square of a sum over M_k. The coefficient of variation

$$\frac{\text{Std}(\Delta S_k)}{\mathbb{E}[\Delta S_k]} = \sqrt{\frac{1 - p_k}{M_k \, p_k}} \xrightarrow{M_k \to \infty} 0.$$

The fluid limit is not an imposed assumption but a consequence of scale: traffic $M_k = N_k \Delta t$ is enormous (tens of thousands of impressions per window for top campaigns), so relative variance decays as $1/M_k$. Hence V_n is negligible against the bias term and is dropped:

$$\min_{\{\mu_k\}} \sum_n \frac{1}{(S_n^*)^2} \big(A_n - S_n^*\big)^2, \qquad A_n = \sum_{k \le n} N_k \Delta t \, c_k \, p_k.$$

7 Derivative, First-Order Condition, Closed Form

7.1 First-Order Condition

Let $J = \sum_n \frac{1}{(S_n^*)^2} (A_n - S_n^*)^2$. The variable μ_k appears only in p_k, which appears in every A_n with $n \ge k$:

$$\frac{\partial J}{\partial \mu_k} = N_k \Delta t \, c_k \frac{\partial p_k}{\partial \mu_k} \cdot \sum_{n \ge k} \frac{2(A_n - S_n^*)}{(S_n^*)^2} \stackrel{!}{=} 0.$$

Since $N_k \Delta t\, c_k \neq 0$ and the win rate is monotone in μ ($\partial p_k / \partial \mu_k \neq 0$),

$$\sum_{n \geq k} \frac{A_n - S_n^*}{(S_n^*)^2} = 0 \qquad \forall k.$$

7.2 Telescoping

Subtracting the equations for k and $k{+}1$ leaves only the $n = k$ term:

$$\frac{A_k - S_k^*}{(S_k^*)^2} = 0 \quad \Longrightarrow \quad \boxed{A_k = S_k^* \quad \forall k.}$$

The pivot. In the fluid limit, optimal pacing means making **expected cumulative spend equal the target curve pointwise**—not "stay close" but "match exactly" (with variance gone, the minimum of a sum of squares is 0 iff every term vanishes).

Telescoping $A_k = S_k^*$ once more gives the **rate equation**:

$$N_k \Delta t\, c_k\, p_k = S_k^* - S_{k-1}^* = \Delta S_k^* \quad \Longrightarrow \quad p_k = \frac{\Delta S_k^*}{N_k \Delta t\, c_k}.$$

In the continuous limit, $N(t)\, P(\mu(t))\, \bar{c}(t) = \dot{S}^*(t)$—"expected spend rate = target rate."

7.3 Solving for the Multiplier

Let the opponents' bid (auction) have CDF F; then $p_k = P(\text{auction}_k < \mu_k v_k) = F(\mu_k v_k)$. Inverting,

$$\boxed{\mu_k = \frac{1}{v_k}\, F^{-1}\!\left(\frac{\Delta S_k^*}{N_k \Delta t\, c_k}\right).}$$

Intuition checks (all consistent):

- Target rate $\Delta S_k^* \uparrow \Rightarrow \mu_k \uparrow$ (want to spend more $\to$ bid harder);
- Traffic $N_k \uparrow \Rightarrow \mu_k \downarrow$ (more opportunities $\to$ no need to push as hard);
- Charge $c_k \uparrow \Rightarrow \mu_k \downarrow$ (each win pricier $\to$ winning fewer suffices).

7.4 Explicit Form with a Logistic Win Rate

With a logistic win rate $p_k = \big(1 + e^{-(\mu_k v_k - b)}\big)^{-1}$ ($\mathbb{R} \to (0,1)$, invertible),

$$\mu_k = \frac{1}{v_k}\left(b + \ln \frac{p_k}{1 - p_k}\right), \qquad p_k = \frac{\Delta S_k^*}{N_k \Delta t\, c_k},$$

yielding a purely elementary closed form. (Modeling the auction with a Beta distribution is more realistic in shape, but its CDF inverse $I_x^{-1}(\alpha, \beta)$ has no elementary expression and must be inverted numerically.)

8 Multiple Advertisers and the Platform

8.1 Formalization

Let advertisers $j \in \{1, \dots, J\}$ each have budget $B^{(j)}$, target curve $S^{*(j)}$, multiplier $\mu^{(j)}$, value $v_k^{(j)}$. Decisions are the allocation $x_k^{(j)} \in \{0, 1\}$ and the multiplier $\mu^{(j)} \geq 0$.

Allocation (auction): impression k goes to the highest paced bid,

$$x_k^{(j)} = 1 \iff j = \arg\max_i \ \mu^{(i)} v_k^{(i)}, \qquad \sum_j x_k^{(j)} \le 1.$$

Second-price billing: winner j is charged

$$c_k^{(j)} = \frac{\max_{i \neq j} \mu^{(i)} v_k^{(i)}}{\mu^{(j)}} \quad \text{(second-highest paced bid, mapped back to } j\text{'s units)}.$$

Advertiser j's problem (given others' $\mu^{(i)},\ i \neq j$):

$$\max_{\mu^{(j)}} \sum_k v_k^{(j)} x_k^{(j)} \quad \text{s.t.} \quad \sum_k c_k^{(j)} x_k^{(j)} \le B^{(j)}.$$

Platform's problem (mechanism designer):

$$\max_{\{x\}} \ R = \sum_k \sum_j c_k^{(j)} x_k^{(j)}.$$

Coupling. Both $x_k^{(j)}$ and $c_k^{(j)}$ contain $\max_{i \neq j} \mu^{(i)} v^{(i)}$—**each advertiser's spend depends on all others' multipliers**. The objective can no longer be minimized per j; the problem rises from single-agent optimal control to a **multi-agent game** + **general equilibrium**.

8.2 When Decoupling Holds

- **Price taker (the common case)**: a single campaign's traffic share is tiny and its perturbation to the overall auction price negligible, so auction$_k$ is treated as an exogenous distribution and **each runs an independent PI controller**—just as a small firm in perfect competition takes the price as given.
- **Price maker (strong coupling)**: a top campaign holds the bulk of some segment's traffic; adjusting its μ moves that segment's price, so it must account for its own feedback on the auction; likewise under scarce, high-value traffic.

9 Equilibrium: Nash, Competitive, Stackelberg

The steady state of the whole advertising system is a layered equilibrium:

1. **Among advertisers (given platform rules)**: the $\mu^{(j)}$ are mutual best responses, none wishing to deviate unilaterally $\Rightarrow$ **Nash equilibrium**. The budget-constrained version is a **pacing equilibrium**.
2. **Plus market clearing** (all impressions sold, all budgets spent, prices equating supply and demand) $\Rightarrow$ **competitive / market equilibrium**. It is stronger than Nash: not only does no one deviate, the market also clears.
3. **Plus the platform choosing rules**: the platform moves first to set the mechanism (GSP/VCG/reserve), advertisers respond $\Rightarrow$ a **Stackelberg** structure.

Equilibrium $\Longleftrightarrow$ dual solution of a global program. Although J advertisers each play and the platform separately maximizes, the point they collectively converge to is exactly the KKT solution of a **centralized convex program** (Eisenberg–Gale type); each advertiser's $\mu^{(j)} = \lambda^{(j)}$ is the dual variable of its budget constraint, i.e. the **market-clearing shadow price**. The decentralized online game implicitly solves one global program (distributed dual ascent).

10 Unification: DP, RL, PI Control

The full problem (stochastic arrivals, unknown distributions, the max operator, game coupling, budget constraints) has **no closed form** and admits only an online numerical solution. Its theoretical parent is a stochastic dynamic program:

$$V_k(S_k) = \min_{\mu_k} \Big\{ w_k(S_k - S_k^*)^2 + \mathbb{E}\big[V_{k+1}(S_{k+1})\big] \Big\}.$$

- **Stochastic DP / Bellman**: the theoretical optimum μ^* is the Bellman solution—but it needs the transition distribution (future traffic, opponents' bids), and with continuous state and many stages, exact solution is infeasible.
- **RL / stochastic approximation**: approximates this DP when model-free. Each noisy sample updates μ; Robbins–Monro guarantees convergence—an "online iterative" extension of the law of large numbers.
- **PI feedback control (the engineering landing)**: the minimal, deployable form of the RL idea. The fixed point it converges to, μ^*, is exactly the dual variable λ^* of the budget constraint.

$$\underbrace{\text{Stochastic DP / Bellman}}_{\text{optimal, needs model}} \xrightarrow{\text{model-free}} \underbrace{\text{RL / stochastic approx.}}_{\text{online estimate of } \lambda} \xrightarrow{\text{engineering}} \underbrace{\text{PI controller}}_{\mu_t \approx \lambda_t}$$

10.1 The Online PI Control Law

Define the normalized error $e_t = (S(t) - S^*(t))/S^*(t)$ ($e_t > 0$: ahead / overspending, approaching the cap, dangerous; $e_t < 0$: behind / underspending):

$$\boxed{\mu_t = \mathrm{clip}\left(\mu_0 + k_p(-e_t) + k_i \sum_{\tau \le t}(-e_\tau),\ 0,\ \mu_{\max}\right).}$$

- **Proportional (P)**: too slow $\Rightarrow \mu\uparrow$ (open); too fast $\Rightarrow \mu\downarrow$ (suppress).
- **Steady-state error**: pure P, under a persistent disturbance, settles at $e_\infty = \dfrac{-\delta}{1+\beta k_p} \neq 0$; finite gain cannot drive it to 0.
- **Integral (I)**: accumulates historical error to drive it to zero (the discrete form of dual ascent).
- **Anti-windup**: freeze the integral when clipping is active, to prevent saturation.
- **Asymmetry**: suppress harder when ahead than when behind ($k_{\text{fast}} > k_{\text{slow}}$), since being ahead breaches the cap—the cost-sensitive asymmetry between the two error types.

10.2 The Boundary of Closed-Form Solvability

Assumption added	Analyticity gained
Fluid limit ($M_k \to \infty$)	drop variance; win rate smooth, differentiable
Single advertiser (no game)	single knapsack, $\lambda^* = v_k/c_k$ in closed form
Price taker	$\max_{i\neq j}$ exogenous, problem decouples
Specified value/auction distribution	win rate F explicit and invertible
Steady state (no transient)	equilibrium reduces to algebraic equations

The full model admits only a numerical solution; a closed form exists but requires stacking the assumptions above, each buying a bit of analyticity at the cost of discarding a dimension of reality (stochasticity, game, market power). **The closed form is a toy for understanding; production uses the numerical online solution (PI / dual ascent).** The former (λ^*) tells you "where the target is"; the latter tells you "how to get there without knowing the future"—and the fixed point the PI controller converges to is exactly the λ^* the closed form provides.

11 Conclusion

This note carried budget pacing from a single advertiser's stochastic optimization all the way to market equilibrium, unifying the dynamic-programming, reinforcement-learning, and feedback-control viewpoints:

- **Single advertiser**: full objective (bias^2 + variance) $\to$ LLN kills variance $\to$ differentiate for the first-order condition $\to$ telescope to $A_k = S_k^*$ $\to$ invert for the closed form $\mu_k = \frac{1}{v_k} F^{-1}\big(\frac{\Delta S_k^*}{N_k \Delta t\, c_k}\big)$.
- **Dual essence**: μ is the Lagrangian dual variable / shadow price of the budget constraint, equal offline to the critical item's bang-per-buck v_k/c_k.
- **Multiple advertisers**: coupled through the auction, the steady state is a competitive equilibrium (with Nash and Stackelberg layers), equivalent to the dual solution of a global convex program.
- **Engineering landing**: the full problem has no closed form, degenerates to RL / stochastic approximation, and is simplified to a PI controller with anti-windup and asymmetric gains.

From a single advertiser's stochastic spend objective to market equilibrium, the pacing multiplier is throughout the shadow price of the budget constraint. The closed form characterizes the target; the online controller reaches it without knowing the future.